

Compare and contrast microcontrollers and single-board computers

Feature	Microcontroller (e.g., Arduino, ESP32)	Single-Board Computer (e.g., Raspberry Pi, NVIDIA Jetson)
Definition	Small integrated chip designed to perform specific tasks	Full computer built on a single board
Operating System	No OS (runs simple firmware)	Full OS (Linux, Windows, Android).
Processing Power	Low (MHz range), optimized for simple tasks.	High (GHz range), multi-core processors.
Memory (RAM/Storage)	Very limited (KB to MB).	Significant (GB of RAM, SD/SSD storage).
Power Consumption	Very low (can run on batteries for months).	High (requires a steady power supply).
Boot Time	Instantaneous.	Takes time to boot the OS.
Connectivity	Basic (Pins for sensors, some have Wi-Fi).	Advanced (USB, HDMI, Ethernet, Wi-Fi).
Use Cases	Embedded systems (sensors, robots, IoT)	Complex applications (AI, web servers, multimedia)

Why use a Microcontroller (MCU) instead of a Single-Board Computer (SBC)?

1. **Power Efficiency is Critical:** If you are building a device that needs to run on a small battery or solar power for a long time (like a remote weather sensor), an MCU is the better choice because it draws significantly less current.
2. **Real-Time Hardware Control:** MCUs are excellent for tasks that require precise timing with hardware, such as reading a sensor at exact intervals or controlling a motor, because there is no operating system "background noise" to cause delays.
3. Cheaper and simpler for small tasks.

Why use a Single-Board Computer (SBC) instead of a Microcontroller (MCU)?

1. **Heavy Processing is Required:** If your project involves complex tasks like computer vision (processing video in real-time), machine learning, or running a web server, the high-speed CPU and RAM of an SBC are necessary.
2. **Multitasking & GUI:** If your project needs to do many things at once - like playing music while displaying a high-resolution user interface on a monitor - an SBC's operating system manages these simultaneous tasks efficiently.
3. Easier to develop with (Python, GUI apps, etc.).

